

Table J-3. Summary of Modified Race and Census 2010 Race Distributions for the United States

Subject	Modified Race		Census 2010	
	Number	Percent	Number	Percent
TOTAL POPULATION	308,745,538	100.0	308,745,538	100.0
One race	299,736,465	97.1	301,761,343	97.7
.Specified race only	280,629,097	90.9	301,761,343	97.7
..White	223,553,265	72.4	241,937,061	78.4
..Black or African American	38,929,319	12.6	40,250,635	13.0
..American Indian and Alaska Native	2,932,248	0.9	3,739,506	1.2
..Asian	14,674,252	4.8	15,159,516	4.9
..Native Hawaiian and Other Pacific Islander	540,013	0.2	674,625	0.2
.Non-specified race only	19,107,368	6.2	(X)	(X)
Two races	8,265,318	2.7	6,346,533	2.1
.Specified race only	5,800,628	1.9	6,346,533	2.1
.Specified and non-specified races	2,464,690	0.8	(X)	(X)
Three or more races	743,755	0.2	637,662	0.2
.Specified race only	567,729	0.2	637,662	0.2
.Specified and non-specified races	176,026	0.1	(X)	(X)
HISPANIC OR LATINO AND RACE				
One race	47,435,002	15.4	49,097,875	15.9
.Specified race only	28,931,899	9.4	49,097,875	15.9
..White	26,735,713	8.7	44,618,105	14.5
..Black or African American	1,243,471	0.4	2,328,113	0.8
..American Indian and Alaska Native	685,150	0.2	1,476,248	0.5
..Asian	209,128	0.1	498,000	0.2
..Native Hawaiian and Other Pacific Islander	58,437	0.0	177,409	0.1
Non-specified race only	18,503,103	6.0	(X)	(X)
Two races	2,799,637	0.9	1,201,228	0.4
.Specified race only	713,760	0.2	1,201,228	0.4
.Specified and non-specified races	2,085,877	0.7	(X)	(X)
Three or more races	242,955	0.1	178,491	0.1
.Specified race only	117,717	0.0	178,491	0.1
.Specified and non-specified races	125,238	0.0	(X)	(X)

NOT HISPANIC OR LATINO AND RACE				
One race	252,301,463	81.7	252,663,468	81.8
..Specified race only	251,697,198	81.5	252,663,468	81.8
..White	196,817,552	63.7	197,318,956	63.9
..Black or African American	37,685,848	12.2	37,922,522	12.3
..American Indian and Alaska Native	2,247,098	0.7	2,263,258	0.7
..Asian	14,465,124	4.7	14,661,516	4.7
..Native Hawaiian and Other Pacific Islander	481,576	0.2	497,216	0.2
Non-specified race only	604,265	0.2	(X)	(X)
Two races	5,465,681	1.8	5,145,305	1.7
..Specified race only	5,086,868	1.6	5,145,305	1.7
..Specified and non-specified races	378,813	0.1	(X)	(X)
Three or more races	500,800	0.2	459,171	0.1
..Specified race only	450,012	0.1	459,171	0.1
..Specified and non-specified races	50,788	0.0	(X)	(X)
(X) Not applicable.				

* Source: U.S. Census Bureau, Population Division

J.1.2.2 County-Level Forecasts

Woods & Poole (2015) developed county-level forecasts for each year from 2000 through 2050, & 2055, by age and gender for non-Hispanic White, African-American, Asian-American, and Native-American and for all Hispanics. The detailed documentation can be found at <http://www.woodsandpoole.com/pdfs/CED15.pdf>. As discussed below, the adjustments necessary to prepare the data for use in BenMAP are relatively straightforward.

For each non-Hispanic subset of the population and each year from 2000-2050 & 2055, we divided the Woods and Poole population for that year by the Woods and Poole population for that subset in 2000. These serve as the growth coefficients for the non-Hispanic subsets of each race. We used a similar calculation to determine the growth rates for the Hispanic population. We assume that each Hispanic race grows at the same rate, and use these growth rates for the Hispanic subsets of each race.⁴³

⁴³ Previous versions of the BenMAP-CE program used a different methodology whereby population estimates for 2000 – 2009 were adjusted using the ratio of 2000 Woods & Poole estimated population and 2000 Census population.

Matching Age Groups Used in BenMAP

There are 86 age groups, so it is a simple matter of aggregating age groups to match the 19 used in BenMAP.

Matching Counties Used in U.S. Census

The county geographic boundaries used by Woods & Poole are somewhat more aggregated than the county definitions used in the 2020 Census and those in BenMAP, and the FIPS codes used by Woods and Poole are not always the standard codes used in the Census. To make the Woods and Poole data consistent with the county definitions in BenMAP, we disaggregated the Woods and Poole data and changed some of the FIPS codes to match the U.S. Census.

Calculating Growth Ratios with Zero Population in 2020

There are a small number of cases where the 2020 county population for a specific demographic group is zero, so the ratio of any future year to the year 2020 data is undefined. In these relatively rare cases, we prepared statewide and national totals and used ratios at the higher levels of geographic aggregation when the more local ratios caused divide-by-zero errors.

J.1.3 PopGrid

If the geographic center of a Census block falls within a population grid-cell, PopGrid assigns the block population to this particular population grid-cell. Note that the grid-cells in an air quality model, such as CMAQ, may cross multiple county boundaries. PopGrid keeps track of the total number of people in each race-ethnic group by county within a particular population grid-cell. Of course, when the population grid-cell is for U.S. counties, then there is only a single county associated with the population grid-cell. However, with air quality models, there can clearly be multiple counties in a population grid-cell.

Keeping track of the total number of people in a county is necessary when forecasting population, as the population forecast for a given grid cell is equal to the year 2020 population estimate from the Census Bureau multiplied by the ratio of future-year to year 2020 county population estimates from Woods & Poole. BenMAP assumes that all age-gender groups within a given race-ethnic group have the same geographic distribution.

To get started with PopGrid, download the program on U.S. EPA's BenMAP GitHub organization and follow the provided instructions, available here:

<https://github.com/BenMAPCE/PopGrid2/tree/main/Vignettes>.

J.1.3.1 PopGrid Output

PopGrid generates two files. One file has the number of people in each grid cell for each of the 304 race-ethnicity-gender-age demographic groups available in PopGrid. Table J-

4 presents an example of what the population file looks like from PopGrid. The Row and Column uniquely identify each grid cell. Note that the Race, Ethnicity, Gender and AgeRange variables are precisely defined (see section on loading population data LoadData_Setups_Population).

Table J-4. Population File Fragment from PopGrid

Row	Column	Year	Population	Race	Ethnicity	Gender	AgeRange
58	81	2000	1.54	WHITE	HISPANIC	MALE	0T00
58	81	2000	0.03	BLACK	HISPANIC	MALE	0T00
58	81	2000	0.01	NATAMER	HISPANIC	MALE	0T00
58	81	2000	0.01	ASIAN	HISPANIC	MALE	0T00
58	81	2000	4.86	WHITE	HISPANIC	MALE	1T04
58	81	2000	0.12	BLACK	HISPANIC	MALE	1T04
58	81	2000	0.03	NATAMER	HISPANIC	MALE	1T04
58	81	2000	0.03	ASIAN	HISPANIC	MALE	1T04
58	81	2000	6.79	WHITE	HISPANIC	MALE	5T09
58	81	2000	0.21	BLACK	HISPANIC	MALE	5T09
58	81	2000	0.05	NATAMER	HISPANIC	MALE	5T09
58	81	2000	0.05	ASIAN	HISPANIC	MALE	5T09
58	81	2000	0.90	WHITE	HISPANIC	MALE	10T014
58	81	2000	0.04	BLACK	HISPANIC	MALE	10T014
58	81	2000	0.01	NATAMER	HISPANIC	MALE	10T014
58	81	2000	0.01	ASIAN	HISPANIC	MALE	10T014
58	81	2000	3.44	WHITE	HISPANIC	MALE	15T019
58	81	2000	0.15	BLACK	HISPANIC	MALE	15T019
58	81	2000	0.04	NATAMER	HISPANIC	MALE	15T019
58	81	2000	0.04	ASIAN	HISPANIC	MALE	15T019
58	81	2000	1.49	WHITE	HISPANIC	MALE	20T024
58	81	2000	0.06	BLACK	HISPANIC	MALE	20T024
58	81	2000	0.02	NATAMER	HISPANIC	MALE	20T024
58	81	2000	0.03	ASIAN	HISPANIC	MALE	20T024
58	81	2000	1.93	WHITE	HISPANIC	MALE	25T029
58	81	2000	0.04	BLACK	HISPANIC	MALE	25T029
58	81	2000	0.01	NATAMER	HISPANIC	MALE	25T029
58	81	2000	0.01	ASIAN	HISPANIC	MALE	25T029
58	81	2000	1.87	WHITE	HISPANIC	MALE	30T0 ₃ 4
58	81	2000	0.08	BLACK	HISPANIC	MALE	30T0 ₃ 4

PopGrid generates a second file that keeps track of the fraction of the total population in each of the eight race-ethnic groups that comes from each county in the United States. Table J-5 presents a sample. The SourceCol and SourceRow uniquely identify each county, and the TargetCol and TargetRow uniquely identify each grid cell. The Value